

Logistic Model 1, weigh for taking alternativ therapy, numerator**The LOGISTIC Procedure**

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	7.5178	0.3817	387.8438	<.0001
trt	0	1	-0.0860	0.0410	4.4009	0.0359
REGION	EUROPE	1	-0.0384	0.0593	0.4196	0.5171
REGION	NORTH AMERICA	1	-0.2966	0.0646	21.0518	<.0001
PANDROG	N	1	0.1546	0.0983	2.4726	0.1158
RACECAT	Non-White	1	0.2416	0.1359	3.1617	0.0754
RACECAT	Unknown	1	-0.6219	0.2098	8.7888	0.0030
ECOG		1	0.1493	0.0908	2.7048	0.1000
month		1	0.3471	0.0338	105.5821	<.0001
month2		1	-2.7194	0.2197	153.2116	<.0001
ALKCAT	<=175	1	0.0990	0.0528	3.5120	0.0609
LDHCAT	<=225	1	0.1687	0.0518	10.5847	0.0011
VISCERAL	N	1	0.1084	0.0539	4.0476	0.0442
ACTHFL	N	1	-0.3295	0.1344	6.0157	0.0142
ANTINEFL	N	1	0.1285	0.0701	3.3596	0.0668

Odds Ratio Estimates			
Effect		Point Estimate	95% Wald Confidence Limits
trt	0 vs 1	0.842	0.717 0.989
REGION	EUROPE vs REST OF WORLD	0.688	0.545 0.870
REGION	NORTH AMERICA vs REST OF WORLD	0.532	0.414 0.683
PANDROG	N vs Y	1.362	0.927 2.003
RACECAT	Non-White vs White	0.871	0.668 1.135
RACECAT	Unknown vs White	0.367	0.201 0.671
ECOG		1.161	0.972 1.387
month		1.415	1.324 1.512
month2		0.066	0.043 0.101
ALKCAT	<=175 vs >175	1.219	0.991 1.499
LDHCAT	<=225 vs >225	1.401	1.144 1.717
VISCERAL	N vs Y	1.242	1.006 1.534
ACTHFL	N vs Y	0.517	0.306 0.876
ANTINEFL	N vs Y	1.293	0.982 1.702

Logistic Model 2, weigh for taking alternativ therapy, denominator**The LOGISTIC Procedure**

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	23.8776	6.2611	14.5438	0.0001
trt	0	1	0.0335	0.0428	0.6130	0.4337
REGION	EUROPE	1	0.00966	0.0614	0.0248	0.8750
REGION	NORTH AMERICA	1	-0.4805	0.0684	49.3291	<.0001
PANDROG	N	1	0.1432	0.1010	2.0120	0.1561
RACECAT	Non-White	1	0.1282	0.1430	0.8038	0.3700
RACECAT	Unknown	1	-0.5495	0.2210	6.1847	0.0129
ECOG		1	0.2479	0.1007	6.0627	0.0138
pd		1	-1.4756	0.0963	234.6658	<.0001
highpsa		1	-1.0654	0.1265	70.8946	<.0001
month		1	0.2297	0.0339	45.7985	<.0001
ecogt		1	-0.1123	0.0591	3.6155	0.0572
psa		1	0.000322	0.000153	4.4067	0.0358
hgb		1	0.1583	0.0530	8.9289	0.0028
hgb2		1	-3.2875	1.1526	8.1351	0.0043
psa2		1	-0.0211	0.00933	5.1120	0.0238
month2		1	-1.5819	0.2280	48.1496	<.0001
ALKCAT	<=175	1	-0.0526	0.0565	0.8684	0.3514
LDHCAT	<=225	1	0.0785	0.0545	2.0714	0.1501
VISCERAL	N	1	0.0125	0.0561	0.0499	0.8232
ACTHFL	N	1	-0.2966	0.1376	4.6443	0.0312
ANTINEFL	N	1	0.1411	0.0721	3.8288	0.0504

Odds Ratio Estimates			
Effect		Point Estimate	95% Wald Confidence Limits
trt	0 vs 1	1.069	0.904 1.265
REGION	EUROPE vs REST OF WORLD	0.631	0.495 0.803
REGION	NORTH AMERICA vs REST OF WORLD	0.386	0.297 0.502
PANDROG	N vs Y	1.332	0.896 1.978
RACECAT	Non-White vs White	0.746	0.565 0.984
RACECAT	Unknown vs White	0.379	0.201 0.715
ECOG		1.281	1.052 1.561
pd		0.229	0.189 0.276
highpsa		0.345	0.269 0.442

Logistic Model 2, weigh for taking alternativ therapy, denominator**The LOGISTIC Procedure**

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
month	1.258	1.177	1.345
ecogt	0.894	0.796	1.003
psa	1.000	1.000	1.001
hgb	1.172	1.056	1.300
hgb2	0.037	0.004	0.358
psa2	0.979	0.961	0.997
month2	0.206	0.132	0.321
ALKCAT <=175 vs >175	0.900	0.721	1.123
LDHCAT <=225 vs >225	1.170	0.945	1.449
VISCERAL N vs Y	1.025	0.823	1.278
ACTHFL N vs Y	0.553	0.322	0.948
ANTINEFL N vs Y	1.326	1.000	1.759

Logistic Model 3 , weight for non-informative censoring, numerator**The LOGISTIC Procedure**

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	5.8940	0.3615	265.8419	<.0001
trt	0	1	0.0476	0.0442	1.1640	0.2806
altRx		1	-1.4342	0.0935	235.3806	<.0001
REGION	EUROPE	1	-0.1379	0.0630	4.7897	0.0286
REGION	NORTH AMERICA	1	0.4922	0.0782	39.6606	<.0001
PANDROG	N	1	-0.0427	0.0889	0.2309	0.6309
ECOG		1	-0.0190	0.0978	0.0377	0.8460
RACECAT	Non-White	1	-0.3033	0.1761	2.9679	0.0849
RACECAT	Unknown	1	0.4487	0.3025	2.2005	0.1380
month		1	-0.3413	0.0252	183.7474	<.0001
month2		1	0.7707	0.1763	19.1166	<.0001
ALKCAT	<=175	1	-0.0162	0.0624	0.0669	0.7959
LDHCAT	<=225	1	0.0673	0.0600	1.2552	0.2626
VISCERAL	N	1	0.0572	0.0632	0.8188	0.3655
ACTHFL	N	1	-0.2487	0.1120	4.9258	0.0265
ANTINEFL	N	1	0.0361	0.0703	0.2644	0.6071

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
trt 0 vs 1	1.100	0.925	1.308
altRx	0.238	0.198	0.286
REGION EUROPE vs REST OF WORLD	1.241	0.988	1.561
REGION NORTH AMERICA vs REST OF WORLD	2.331	1.766	3.077
PANDROG N vs Y	0.918	0.648	1.301
ECOG	0.981	0.810	1.188
RACECAT Non-White vs White	0.854	0.648	1.125
RACECAT Unknown vs White	1.811	0.752	4.364
month	0.711	0.677	0.747
month2	2.161	1.530	3.053
ALKCAT <=175 vs >175	0.968	0.758	1.237
LDHCAT <=225 vs >225	1.144	0.904	1.448
VISCERAL N vs Y	1.121	0.875	1.436
ACTHFL N vs Y	0.608	0.392	0.944
ANTINEFL N vs Y	1.075	0.816	1.416

Logistic Model 4 weight for non-informative censoring, denominator**The LOGISTIC Procedure**

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	8.1984	6.5668	1.5586	0.2119
trt	0	1	0.0337	0.0447	0.5681	0.4510
altRx		1	-1.4982	0.0996	226.0778	<.0001
REGION	EUROPE	1	-0.1599	0.0638	6.2815	0.0122
REGION	NORTH AMERICA	1	0.5165	0.0804	41.2848	<.0001
PANDROG	N	1	-0.0180	0.0900	0.0398	0.8419
RACECAT	Non-White	1	-0.3016	0.1773	2.8955	0.0888
RACECAT	Unknown	1	0.4412	0.3039	2.1075	0.1466
ECOG		1	-0.0547	0.1089	0.2525	0.6153
pd		1	0.0513	0.1002	0.2619	0.6089
highpsa		1	0.5330	0.1194	19.9229	<.0001
month		1	-0.3288	0.0257	163.3943	<.0001
ecogt		1	0.0846	0.0681	1.5429	0.2142
psa		1	0.000025	0.000138	0.0332	0.8554
hgb		1	0.0107	0.0544	0.0385	0.8445
hgb2		1	-0.3039	1.1962	0.0645	0.7995
psa2		1	-0.0156	0.00921	2.8675	0.0904
month2		1	0.5943	0.1827	10.5868	0.0011
ALKCAT	<=175	1	-0.0106	0.0663	0.0256	0.8729
LDHCAT	<=225	1	0.0602	0.0607	0.9831	0.3214
VISCERAL	N	1	0.0717	0.0639	1.2571	0.2622
ACTHFL	N	1	-0.2436	0.1125	4.6913	0.0303
ANTINEFL	N	1	0.0263	0.0710	0.1368	0.7115

Odds Ratio Estimates			
Effect		Point Estimate	95% Wald Confidence Limits
trt	0 vs 1	1.070	0.898 1.274
altRx		0.224	0.184 0.272
REGION	EUROPE vs REST OF WORLD	1.217	0.966 1.534
REGION	NORTH AMERICA vs REST OF WORLD	2.394	1.801 3.183
PANDROG	N vs Y	0.965	0.678 1.373
RACECAT	Non-White vs White	0.850	0.643 1.124
RACECAT	Unknown vs White	1.787	0.739 4.323
ECOG		0.947	0.765 1.172

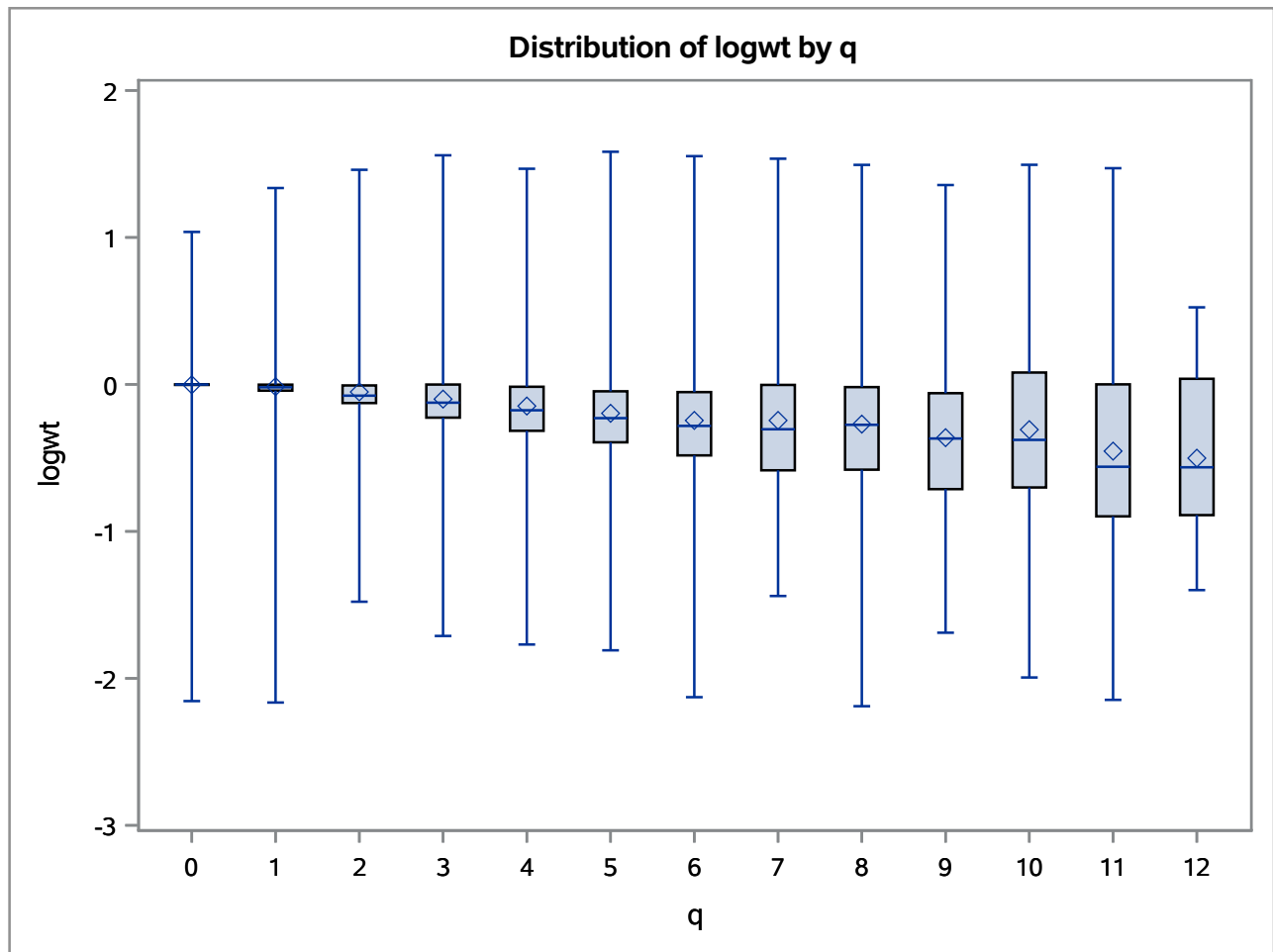
Logistic Model 4 weight for non-informative censoring, denominator**The LOGISTIC Procedure**

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
pd	1.053	0.865	1.281
highpsa	1.704	1.348	2.153
month	0.720	0.684	0.757
ecogt	1.088	0.952	1.244
psa	1.000	1.000	1.000
hgb	1.011	0.908	1.125
hgb2	0.738	0.071	7.696
psa2	0.985	0.967	1.002
month2	1.812	1.267	2.592
ALKCAT <=175 vs >175	0.979	0.755	1.270
LDHCAT <=225 vs >225	1.128	0.889	1.431
VISCERAL N vs Y	1.154	0.898	1.483
ACTHFL N vs Y	0.614	0.395	0.955
ANTINEFL N vs Y	1.054	0.798	1.392

check weight: mean should be near 1, MSM weight

Obs	n	mean	std	min	max
1	17412	0.98023	0.46269	0.11197	20.6946

After rescaling, check weight: mean should be near 1, MSM weight

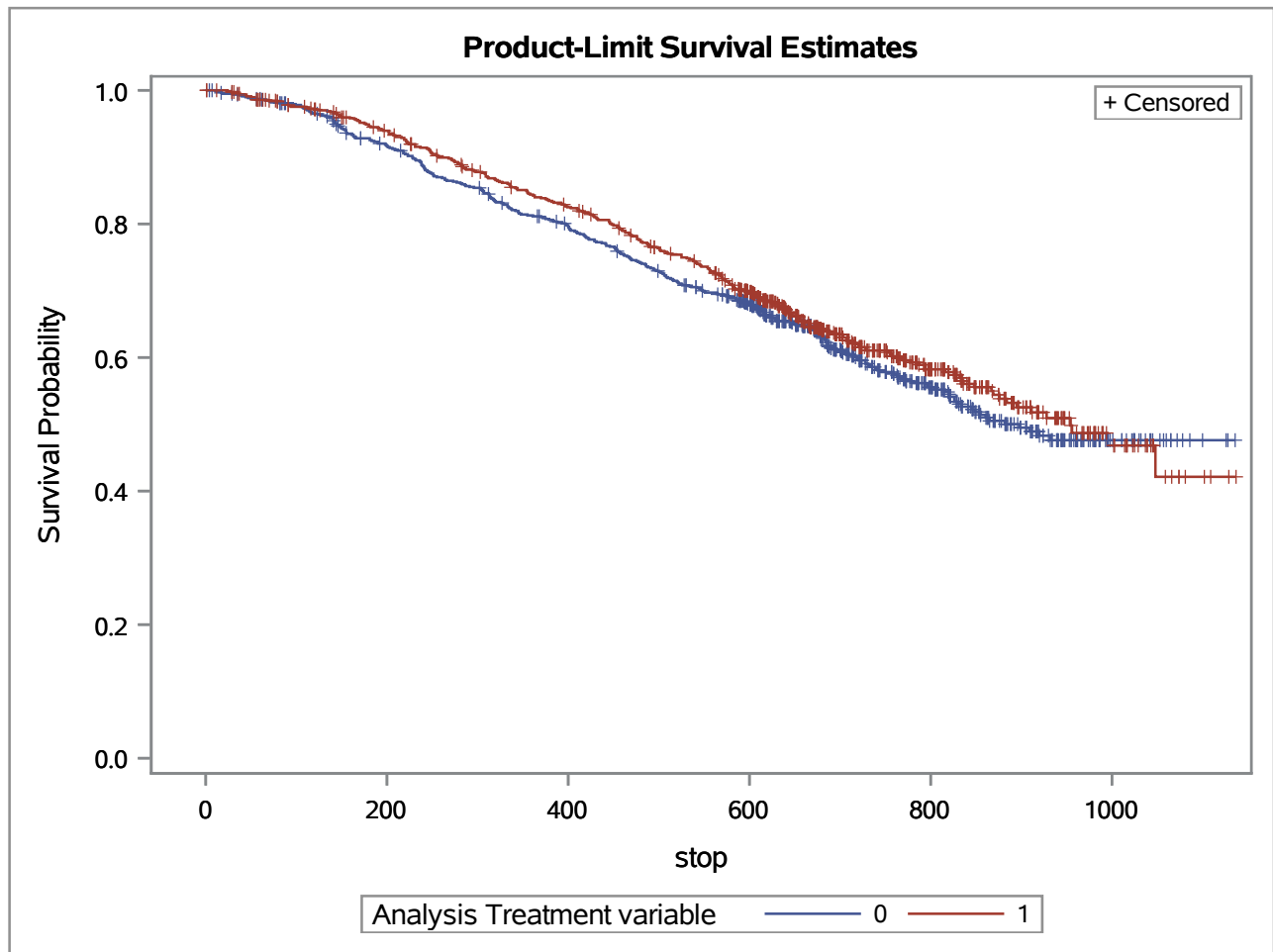


After rescaling, check weight: mean should be near 1, MSM weight

Obs	n	mean	std	min	max
1	17412	0.97100	0.33294	0.11197	4.86935

Log-rank for OS, reproduce ELM-PC4 result
Using proc lifetest, create data for R OS plot

The LIFETEST Procedure



MSM Cox model for OS, Check significance of R by A(t) interaction term Using proc phreg, counting process data format for multiple records

The PHREG Procedure

Joint Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
trt	1	4.9564	0.0260
altRx	1	0.0267	0.8702
altRx*trt	1	0.9080	0.3406

Note: Under full-rank parameterizations, Type 3 effect tests are replaced by joint tests. The joint test for an effect is a test that all of the parameters associated with that effect are zero. Such joint tests might not be equivalent to Type 3 effect tests under GLM parameterization.

MSM Cox model for OS

Using proc phreg, counting process data format for multiple records

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The PHREG Procedure

Type 3 Tests			
Effect	DF	Wald Chi-Square	Pr > ChiSq
trt	1	3.8732	0.0491
altRx	1	0.4108	0.5216

Analysis of Maximum Likelihood Estimates											
Parameter		DF	Parameter Estimate	Standard Error	StdErr Ratio	Chi-Square	Pr > ChiSq	Hazard Ratio	95% Hazard Ratio Confidence Limits		Label
trt	1	1	-0.18526	0.09413	1.185	3.8732	0.0491	0.831	0.691	0.999	Analysis Treatment variable 1
altRx		1	0.06564	0.10242	1.181	0.4108	0.5216	1.068	0.874	1.305	Alternate Therapy Indicator

z-score and p-value from weighted Log-rank test
Analysis is not stratified, adjusted by MSM

Obs	z	pvalue
1	-2.21829	0.026535

Time to OS after Adjusting for Alt. Rx effect
Using MSM Approach

